

Di-ethylhexyl phthalate (DEHP): Technical Report on Conditions of Use

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About this Technical Report

On August 23, 2019, the U.S. Environmental Protection Agency (EPA) released what it describes as “[reasonably available information](#)” about Di-ethylhexyl phthalate (DEHP) and requested comments about the chemical's conditions of use. DEHP is one of twenty proposed high-priority substances for evaluation under the Toxic Substances and Control Act.

In response to EPA's call for such information, this Technical Report provides an overview of the manufacturing, processing, distribution into commerce, and disposal of DEHP. It also provides an accounting of DEHP releases from these conditions of use.

Some of the information in the following report is not in EPA's August 2019 document. While EPA's list of industries that use DEHP is comprehensive, the following technical report provides further details on:

- The leading manufacturers, processors, distributors, and disposers of DEHP.
- Efforts by the medical community to eliminate DEHP from medical devices.
- Releases of DEHP from common consumer goods including PVC floors.
- The transfer of DEHP waste to overseas recycling companies.

The following technical report is based on public information, including company literature and filings; state and federal agency data, especially from the EPA, Centers for Disease Control, and U.S. Customs and Border Protection; and overseas government agencies.

Note: Weights are measured in pounds for consistency with how EPA reports most data.

A. Identifications

Chemical Name: Di-ethylhexyl phthalate (DEHP)

EPA Docket Number: <https://www.regulations.gov/docket?D=EPA-HQ-OPPT-2018-0433>

CAS RN: 117-81-7 (primary). Additional: 109630-52-6; 126639-29-0; 137718-37-7; 15495-94-0; 205180-59-2; 275818-89-8; 40120-69-2; 50885-87-5; 607374-50-5; 8033-53-2.

Synonyms¹:

- n-Dioctyl phthalate
- 1, bis(ethylhexyl) ester
- 1,2-Benzenedicarboxylic acid bis-(1-ethylhexyl) ester
- 1,2-Benzenedicarboxylic acid bis(2-ethylhexyl) ester
- 1,2-Benzenedicarboxylic acid, 1,2-bis(2-ethylhexyl) ester
- 1,2-Benzenedicarboxylic acid, bis(2-ethylhexyl) ester, labeled with carbon-14
- 1,2-bis(2-ethylhexyl) benzene-1,2-dicarboxylate
- 14C -DEHP
- 2-ethylhexyl 2-[(2-ethylhexyl)oxycarbonyl]benzoate
- 2-ethylhexyl phthalate
- Behp
- benzene-1,2-dicarboxylic acid bis(2-ethylhexyl) ester
- Benzenedicarboxylic acid, bis(2-ethylhexyl) ester
- bis (2-Ethylhexyl) Phthalate
- Bis (2-Ethylhexyl) Phthalate (Dioctyl phthalate)
- bis-(2-ethylhexyl) 1,2-benzenedicarboxylate
- Bis(2-ethylhexyl) benzene-1,2-dicarboxylate
- Bis(2-ethylhexyl) o-phthalate
- bis(2-ethylhexyl) phthalate
- Bis(2-ethylhexyl)ester phthalic acid
- Di-2-ethylhexyl phthalate
- Di-sec-octyl phthalate
- Di(2-ethylhexyl) o-phthalate
- Di(2-ethylhexyl) orthophthalate
- Diethylhexylphthalate (Bis-(2-ethylhexyl) Phthalate)
- Dioctyl Phthalate (DOP)
- N-Dioctyl phthalate
- octyl phthalate
- Phthalic acid bis(2-ethylhexyl ester)
- Phthalic acid di(2-ethylhexyl) ester
- Phthalic acid dioctyl ester

B. Manufacturing

Di-ethylhexyl phthalate (DEHP) is manufactured by the reaction of 2-ethylhexanol with phthalic anhydride. The esterification process uses either an acid (typically sulfuric acid or para-toluenesulfonic acid) or metal catalyst or, at high temperature, no catalyst.²

According to EPA, DEHP's "national aggregate production volume" (defined as the combination of domestic manufacturing and imports) ranged between 100 and 250 million pounds per year from 1986 to 2015. EPA provided an exact production volume for one year (2011) of 152,694,720 pounds.³

1. U.S. Manufacturers

Two companies have reported manufacturing DEHP in EPA Chemical Data Reports (CDR).

a. Eastman Chemical Co. (Kingsport, Tennessee)

Eastman reported manufacturing DEHP in its 2012 and 2016 CDR submissions. Eastman reported that the chemical is intended for use – in concentrations of up to 30% of the product's weight – in paints and coatings, and commercial plastic and rubber products.

Eastman has been the country's leading producer of ortho-phthalates, including DEHP, and has developed non-phthalate plasticizers in response to changing market conditions.

However, EPA Toxics Release Inventory (TRI) data does not reflect a decline in the production of DEHP. Release to air and land in 2018 (1,174 pounds) were similar to releases 10 years earlier (1,129 pounds). Eastman has not submitted Pollution Prevention Plans with EPA, nor has it announced a phase-out in DEHP manufacturing.⁴

b. Teknor Apex Company (Brownsville, Tennessee)

In its 2012 CDR, Teknor Apex reported that it manufactured DEHP at its plant in Brownsville, Tennessee. In 2010, Teknor Apex reported production of 3.6 million pounds, and 1.8 million pounds in 2011. EPA is withholding production values for years since 2011.

This factory processes DEHP in the manufacture of medical supplies, garden hoses, and other heavily plasticized products including automotive interiors and

wire insulation.⁵ In 2018, it reduced DEHP pollution by 50% after its technical team “worked to remove DEHP from products.”⁶

C. Importing and Processing

More DEHP is imported than domestically produced. Over 14 million pounds of DEHP were imported between 2012 and November 5, 2019, according to U.S. Customs shipping records in the Panjiva database.

1. Processors

The following companies process imported DEHP, according to their CDR information submitted to EPA:

- a. **Hallstar Company (Chicago, Illinois).** As of 2018, Hallstar is divided into two divisions: industrial and cosmetic, both of which appear to do more processing than distributing. The industrial division’s website says, “Hallstar has been manufacturing polymer additives for over 50 years, and is a market leader in the design and synthesis of polymer modifiers.”⁷
- b. **LG Chem America, Inc. (Atlanta, GA and Cerritos, CA).** LG Chem’s Cerritos electronics plant is “focused on the solar market.”⁸
- c. **Mexichem SAB DE CV – Alphagary Corp. (Worcester, Massachusetts and Mecklenburg, North Carolina).** According to Mexichem’s CDR, 61% of the DEHP in Mecklenburg is incorporated in medical devices, 28% in plastic products, and 1% in “food, beverage and tobacco products.” Mexichem reported exporting relatively small quantities of DEHP from Worcester, and large quantities (over 1 million pounds one year) from Mecklenburg.⁹
- d. **Polyone Corporation (Elyria, Ohio and Terre Haute, Indiana).** Polyone, a PVC resin manufacturer, reported that DEHP is incorporated into furniture, personal care products, building and construction materials, and electrical and electronic products. It also exports small volumes of DEHP.
- e. **Primex Plastics Corporation (Richmond, IN).** Primex is a subsidiary of the distributor ICC Industries (see 2.d., below). Primex manufactures shipping solutions, color concentrates, and extruded plastic sheeting.

2. Distributors

Many importers reporting CDR information on DEHP in 2012 and/or 2016 are chemical distributors, not processors. These include:

- a. **Alac International, Inc.** (New York, New York). According to U.S. shipping records, Alac imported 38,000 pounds of DEHP from **Nan Ya Plastics Corporation (Taiwan)** in 2012.
- b. **Industrial Chemicals, Inc. (Vestavia Hills, Alabama)** This company imported 476,160 pounds of DEHP in 2011, according to its CDR filing. Its suppliers, according to records in the Panjiva trade database, are **Aarti Industries** and **KLJ Plasticizers**, both in India. Since 2012, Industrial Chemicals has imported over 407,000 pounds of DEHP from these companies.

The following ten companies had no Panjiva shipping records that corresponded with their CDR filings:

- c. **ChemSpec, Ltd. (Stark, Ohio)**
- d. **ICC Industries Inc. (New York, New York)**
- e. **Greenchem (Palm Beach, Florida)**
- f. **FRP Services & Co. (America) Inc. (New York, New York)**
- g. **Shrieve Chemical (formerly MC International, LLC) (Miami, Florida)**
- h. **R.E. Carroll, Inc. (Trenton, New Jersey)**
- i. **Silver Fern Chemical, Inc. (Seattle, Washington)**
- j. **The Chemical Company (Jamestown, Rhode Island)**
- k. **Tricon International (Houston, Texas)**
- l. **Univar USA, Inc. (Redmond, Washington)**

3. Importers of More Than 25,000 Pounds Not Filing CDR

In addition to the above companies that reported CDR information to EPA, U.S. trade records published in the Panjiva database identify additional companies that have imported DEHP during one or more of the past reporting years of 2012 to 2015. Several of these companies imported at rates higher than the mandatory reporting threshold of 25,000 pounds per year.

- a. **Innua USA (Wilmington, North Carolina).** Innua is a PVC and plasticizer distributor. It has imported over 2.3 million pounds of DEHP since 2014. Its suppliers include **Nan Ya Plastics Corporation (Taiwan)**, **Aekyung Petrochemical (South Korea)**, **UPC Technology (Taiwan)**, and **Frescom Company (South Korea)**.
- b. **Resin Technology (Groton, Massachusetts).** This company imported 87,212 pounds of DEHP from UPC Technology in 2014.
- c. **V.T. Mancusi (Lawrence, New York).** This company imported 118,928 pounds of DEHP from KLJ Plasticizers (India) between 2013 and 2014.

The following companies imported DEHP only after the last reportable year, 2015.

- d. **Matrix Chemical (Plano, Texas).** Matrix imported 39,000 pounds of DEHP from UPC Technology in 2012.
- e. **Formosa Global Solutions (Livingston, New Jersey).** Formosa imported 1.5 million pounds between 2017 and November 2019 from **Nan Ya Plastics**. Separately, Formosa advertises phthalate-free PVC resins.¹⁰
- f. **Oxyde Chemical (Houston, Texas).** Oxyde imported 219,629 pounds of DEHP from **Oxyde Chemicals China** in 2017. Oxyde is a global petrochemical and plastics trader.¹¹

Through these distribution and importation channels, DEHP is widely distributed to plastics producers around the United States. Toxics Release Inventory air release data identifies the types of facilities that are processing DEHP.

Table 1. DEHP Releases to Air, Total, 2012 to 2018

Processor	Facility Type	Air	% of whole
Honda Manufacturing (Lincoln, AL)	Automobiles	78,335	29%
Teknor Apex (Brownsville, TN)	Plasticizers, PVC compounding	36,058	14%
Chemprene LLC (Beacon, NY)	Rubber coated textiles	13,860	5.2%
Eaton (Newbern, TN)	Hydraulics	13,303	5.0%
Achilles USA Inc (Everett, WA)	Plastic films	12,351	4.6%
Baxter Healthcare (Mountain Home, AR)	Medical supplies	11,452	4.3%
Kia Motors (West Point, GA)	Automobiles	10,125	3.8%
Northwest Pipe Co (Brea, CA)	Engineered steel pipe	7,994	3.0%
Teknor Apex Co (Pawtucket, RI)	PVC compounding	6,356	2.4%
Entegris Inc (Bedford, MA)	Microelectronics coatings	6,156	2.3%
Eastman Chemical Co (Kingsport, TN)	Plasticizers	6,063	2.3%
Eaton Aeroquip Inc (Forest City, NC)	Hydraulics	4,081	1.5%
Teknor Apex - Carolina Co (Fountain Inn, SC)	PVC compounding	3,797	1.4%
Midwest Elastomers Inc (Wapakoneta, OH)	Plastics grinding	3,500	1.3%
Fiskars Brands Inc (Excelsior Springs, MO)	Scissors	3,085	1.2%
Baxter Healthcare (Madison, NC)	Medical supplies	3,015	1.1%
Hallstar Co. (Memphis, TN)	Plasticizers	2,932	1.1%
Pass & Seymour Legrand (Concord, NC)	Electrical devices	2,750	1.0%
American Renolit Corp (La Porte, IN)	Plastic films	2,733	1.0%
LG Hausys America Inc (Adairsville, GA)	Solid surfaces (e.g., countertops)	2,500	0.9%
O'Sullivan Films Inc (Winchester, VA)	Plastic films	2,478	0.9%
American Renolit Corp LA (Commerce, CA)	Plastic films	2,423	0.9%
Ford Motor Company (Claycomo, MO)	Automobiles	2,312	0.9%
Teknor Apex Co (City Of Industry, CA)	PVC compounding	2,135	0.8%
OTHERS		26,431	
TOTAL		292,657	

Source: EPA Toxic Release Inventory Data.

D. Distribution

DEHP is widely distributed in flexible PVC products such as garden hoses, gloves, toys, and medical devices. These include domestic and imported PVC products. Imported PVC products are more likely to contain DEHP than domestically produced PVC. According to the National Institutes of Health in 2005, “DEHP accounted for about 10% of plasticizers used in the U.S. to make products, compared with about 50-60% in Japan and other parts of Asia.”¹²

In 2008, the U.S. Congress passed the Consumer Product Safety Improvement Act (CPSIA), making it unlawful to distribute DEHP at concentrations of >0.1% for use in any children’s toy or item for children’s use. Use of DEHP is also restricted in cosmetics. In 2013, with the passage of California’s Proposition 65, products that contain DEHP must be labeled as such.¹³

Retail stores also have been pressuring suppliers to eliminate ortho-phthalates. In 2015, large chain stores including Home Depot, Lowes, and Lumber Liquidators began to phase out DEHP-containing vinyl flooring, a transition that was largely complete by 2019.¹⁴ In 2019, additional retailers enacted chemicals policies to restrict suppliers’ use of DEHP, including Staples and Ahold Delhaize.¹⁵

1. Medical Devices

According to the Department of Health and Human Services at the National Toxicology Program, “the primary routes of potential human exposure to DEHP are ingestion, inhalation, and through medical procedures; exposure levels are highest for medical procedures.”¹⁶

A large share of DEHP is processed and incorporated in medical devices, such as blood bags, catheters, and oxygen masks. In 2005, according to National Institute of Health, this share was 40% of all DEHP made in the United States.¹⁷ According to a European Union scientific committee report, premature neonates in intensive care units “may receive even higher DEHP exposures than adults relative to their (bodyweight). Such exposures may occur for a period of weeks or even months.”¹⁸

In 2006, a U.S. National Toxicology Program Expert Panel “re-affirmed that the phthalate DEHP poses a risk to human development and fertility,” according to Health Care Without Harm.¹⁹ Hospitals have since demanded alternatives to DEHP-containing plastics in medical devices.

Those demands have not been heard by some of the largest medical equipment suppliers. In addition to exposing patients to DEHP, these companies also are responsible for a vast amount of DEHP waste. From 2012 to 2018, medical product

manufacturers transferred 15.2 million pounds of DEHP waste offsite (see Transfers table below).

Alternatives are in hand and growing in market share: In 2017, at least 168 U.S. hospitals reported purchasing at least one DEHP-free product line. The global market for non-PVC IV bags was estimated at \$1.1 billion and is expected to rise steadily through 2025.²⁰

E. Releases

1. Releases from Manufacturing

The ongoing manufacturing, processing, distribution, recycling, and release of DEHP leads to its release into the environment and people's exposure to it. From 2012 to 2018, more than 150 companies, representing a wide range of industries, reported releasing DEHP into the air, water, or land, and/or transferring DEHP waste to other communities.

In 2012, the Tennessee Department of Environmental Conservation brought suit against Teknor Apex for significant violations of discharge limits of DEHP dating back to 2008. Violations continued in 2017, according to a state inspection.²¹ The Brownsville, Tennessee factory is one of the two DEHP manufacturers in the United States (see B.1.b., above).²²

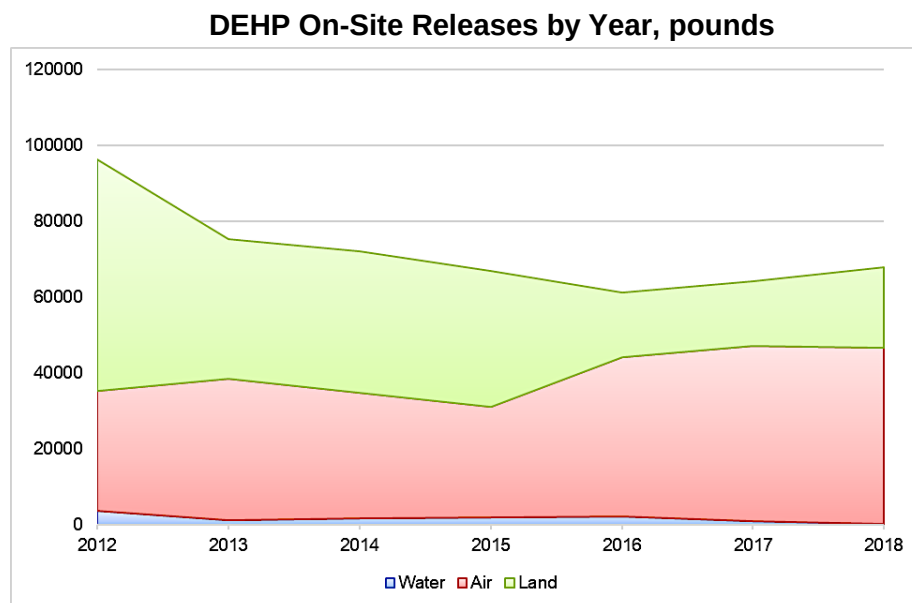


Figure 1. Source: U.S. Toxic Release Inventory Data.

2. Releases from Products

Once in commerce, DEHP continues to be released. DEHP migrates, slowly but consistently, from plastics. According to a Danish study, migration rates change with conditions: they are slowest on simple contact, greater when exposed to sweaty skin, and greatest with prolonged contact with saliva.²³

This has been known for some time: In 2001, FDA found that due to releases, “children undergoing certain medical procedures may represent a population at increased risk for the effects of DEHP.”²⁴ Medical provider Kaiser-Permanente announced in 2012 that it would no longer purchase medical products that contain DEHP. A spokesperson stated,

The issues are out there for both our patients and the environment. PVC and DEHP are both commonly used in IV bags, IV tubing, and product packaging. DEHP is added to make the plastic more flexible; however, it does not adhere to the plastic, but ‘floats’ within the vinyl structure. During certain therapies, DEHP can leach from the structure and be infused along with the IV fluid, medication, or blood transfusion, thus entering the patient's bloodstream. Blood products and several drugs, such as Paclitaxel, enhance leaching of DEHP from PVC containers. IV containers that do not contain DEHP eliminate the risk of toxicity, however. DEHP has been scientifically documented as a reproductive toxicant.²⁵

Children come into contact with DEHP released from a variety of products, including toys and building materials, and flooring and wall coverings. In 2005, researchers in Sweden found DEHP (and butyl benzyl phthalate) concentrations in dust were associated with amounts of PVC flooring and wall materials in their study of 390 homes.²⁶ In 2014, French researchers evaluated indoor air and dust in 30 French homes; again, phthalates, including DEHP, had the highest concentrations in both air and dust.²⁷



Figures 2 and 3. This product, sold at Hobby Lobby in 2015, contained 33% DEHP, the highest concentration of any product yet tested by the State of Washington Department of Ecology.²⁸ Washington prohibits the distribution of children's products containing more than 0.1% DEHP. STATE OF WASHINGTON DEPARTMENT OF ECOLOGY PHOTOS, 2015.

3. Releases from Recycling

Operations that recycle scrap containing DEHP release it into the environment. For example, according to Healthy Building Network,

A study reported in the Bulletin of Environmental Contamination & Toxicology (2009) identified some of the impacts of recycling imported wastes in the Taizhou area. 'Phthalate esters (PAEs) including dimethyl phthalate (DMP), diethyl phthalate (DEP), di-n-butyl phthalate (DBP), Di-(2- ethylhexyl) phthalate (DEHP) and di-n-octyl phthalate (DnOP) in the e-waste soils were collected and analyzed from sites Fengjiang, Nanshan and Meishu in Taizhou city,' reported a local team of scientists. 'The result showed that the total PAEs concentrations ranged from 12.566 to 46.669 mg/kg in these three sites. DEHP, DBP and DEP were the major phthalates accounting for more than 94% of total phthalates studied. Comparing to the results from other studies, the e-waste soils from Taizhou city were severely contaminated with PAEs.'²⁹

Two of the top five destinations for DEHP waste transfers are in other countries. Baxter, the leading distributor of DEHP to hospitals through its medical supplies, also distributed DEHP waste to its subsidiary in Mexico. Plasticos Del Caribe, a plastics packaging company in Santo Domingo, Dominican Republic received the fifth-most DEHP waste.

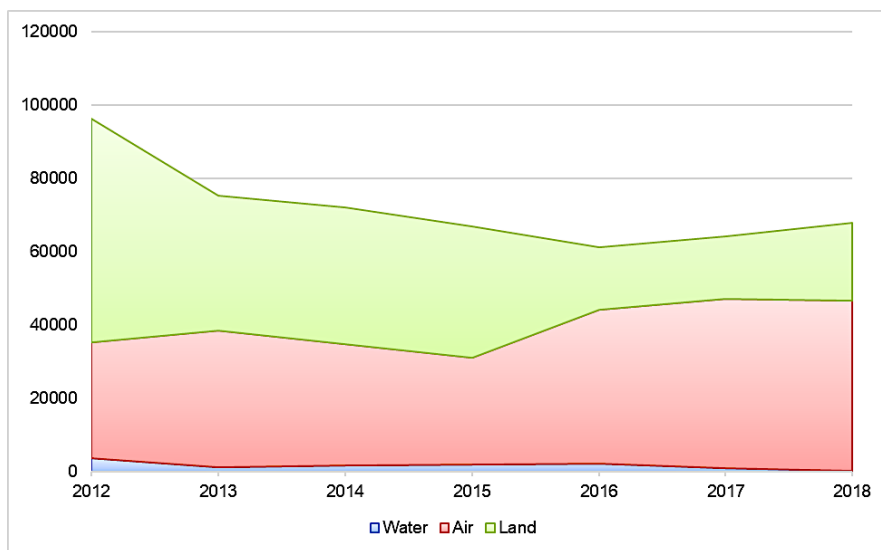


Figure 4. DEHP Releases by Year, pounds, U.S. Source: EPA Toxics Release Inventory

Table. Di-ethylhexyl phthalate imports and releases, U.S., total pounds, 2012 to 2018.

Facility Name	Air	Water	Land	Off-Site Transfer	Sum
ICU Medical Fleet Services (Austin, TX)	94.4	22.28		5,457,889	5,458,005
Baxter Healthcare (Aibonito, PR)	407.4			3,975,875	3,976,282
Baxter Healthcare (Jayuya, PR)	3.35			130,5427	1,305,430
Mgs Buffalo Inc (Buffalo, NY)				1196058	1,196,058
Baxter Healthcare (Cleveland, MS)	236			1183394	1,183,630
ICU Medical Fleet Services (Round Rock, TX)	39.5			1138084	1,138,124
American Renolit Corp (La Porte, IN)	2,733			959751.8	962,485
Entegris Inc (Bedford, MA)	6,156			857,438	863,594
Fiskars Brands Inc (Excelsior Springs, MO)	3,085			635394	638479
Cosmoflex Inc (Hannibal, MO)	123			332830	332953.5
Baxter Healthcare Corp (Marion, NC)	137			326141.2	326278.9
Baxter Healthcare (Mountain Home, AR)	11452		3000	303475.2	317927.2
Contitech USA Inc (Norfolk, NE)	483.4009			279181.5	279664.9
Zimmer Surgical (Dover, OH)	35			253653	253688
Haemonetics Corp (Braintree, MA)				197620	197620
Flexible Technologies Inc (Abbeville, SC)				189379	189379
Teknor Apex Tennessee Co (Aka Haywood Co) (Brownsville, TN)	36,058	44		136050	172152
Hexpol Compounding Colonial Rubber Works				158247	158247
Royal Mouldings LTD (Marion, VA)	198			148113	148311
Midwest Elastomers Inc (Wapakoneta, OH)	3500			144595	148095
Chemical Waste Management Of The Northwest Inc (Arlington, OR)			116076.7	8771.9	124848.6
Kia Motors Manufacturing of Georgia (West Point, GA)	10125			109402	119527

Fenwal International Inc (Maricao, PR)	61.87		115866	115927.9
Carefusion (Ontario, CA)			106251	106251
Sylvin Technologies Inc (Denver, PA)	31.554	1.9896	99583.13	99616.68
Honda Manufacturing of Alabama LLC (Lincoln, AL)	78335		18016	96351
Parker Hannifin Corp Inc (Red Oak, IA)			84397	84397
Wayne Disposal Inc (Belleville, MI)	17.6	81735.18	293.09	82045.87
Contitech USA Inc (Lincoln, NE)	57.1324		75008.73	75065.86
Conmed Corp (Utica, NY)	308.53		73376.2	73684.73
Colorite Plastics Co (Sparks, NV)			71795	71795
Lauren Manufacturing Co (New Philadelphia, OH)			62863	62863
Graniteville Specialty Fabrics (Graniteville, SC)			61050	61050
Baxter Healthcare of Puerto Rico (Denver, PA)			60040	60040
Teknor Apex - Carolina Co (Fountain Inn, SC)	3797	0.31	52077	55874.31
Eaton - Newbern (Newbern, TN)	13302.88		42417.15	55720.03
Michigan Rubber Products Inc (Cadillac, MI)	60		52964.53	53024.53
Nov Fiber Glass Systems-Burkburnett (Burkburnett, TX)	698		50628	51326
Rotadyne Roll Group La Porte (La Porte, IN)	1135.1		47349	48484.1
Mantaine Corp (Mantua, OH)	3.98		46793.6	46797.58
Saint-Gobain Performance Plastics (Akron, OH)	1351	3.95	44716.9	46071.85
Coleman Cable LLC East Facility (Bremen, IN)			45267	45267
Contitech USA Inc (Mount Pleasant, IA)	384		43839.05	44223.05
All-American Hose Acquisition LLC (Union City, PA)			42200	42200
Kirkhill Inc (Brea, CA)			36986	36986
Cooper Standard Automotive (Bowling Green, OH)			34650	34650
Coeur Inc (Sheboygan, WI)	84.4		32467	32551.4
Sumitomo Electrical Wiring Systems Inc (El Paso, TX)			27682	27682
Poly Vinyl Co Inc (Sheboygan Falls, WI)	271.6		27309	27580.6

Parker Hannifin Alliance Plant (Alliance, NE)	1500			26024	27524
Eaton Aeroquip Inc (Forest City, NC)	4081			22894	26975
Loxcreen Co Inc (Brooklet, GA)	834			24589	25423
American Roller Co LLC (Union Grove, WI)				24486	24486
Polymerics Inc (Cuyahoga Falls, OH)				24381	24381
Ronken Industries Inc (Spring Valley, IL)	173.17	3.56		23027.35	23204.08
Rotadyne Roll Group Romeoville (Romeoville, IL)	923.3			21601	22524.3
US Ecology Nevada Inc (Beatty, NV)			22173.57	0.32	22173.89
Univar USA Inc Metro Blvd (Maryland Heights, MO)	87.75			21238.33	21326.08
Eastman Chemical Co (Kingsports, TN)	6063	8072	1240	5093	20468
Chemprene LLC (Beacon, NY)	13860.13			6126.07	19986.2
Teknor Apex Co (City Of Industry, CA)	2135	2625		14295	19055
Rotadyne Roll Group Leola (Leola, PA)	423.3			17319	17742.3
Interfaceflor (Lagrange, GA)	1380			14774.31	16154.31
Nov Wellbore Technologies - DDS - Willis (Willis, TX)				15237.31	15237.31
Profusion Industries LLC (Marietta, OH)	47.974			15159.47	15207.44
Biltrite Ripley Operations LLC (Ripley, MS)	112.17			14867.39	14979.56
Northwest Pipe Co (Brea, CA)	7994.8			6307.8	14302.6
Veka Inc (Fombell, PA)				13749	13749
Elastomer Processing Group - Chicago (Chicago, IL)	202.53			13223	13425.53
Fenner Drives (Manheim, PA)	134			12782	12916
Achilles USA Inc (Everett, WA)	12351				12351
HBD IND Inc (Oneida, TN)				12288	12288
Kohler Co (Union City, TN)				12052	12052
Nov Rig Systems Rubber Plant %26 Controls Building (Houston, TX)				11385	11385
Grace Davison-Edison (Edison, NJ)	83			10618	10701
Teknor Apex Co (Pawtucket, RI)	6355.6			3842.16	10197.76

Chem Technologies LTD (Middlefield, OH)	2051			7925	9976
HBD Industries (Bellefontaine, OH)				9386	9386
Vital Signs, Inc. (Totowa, NJ)	5			9010	9015
Parker Hannifin Manhattan Facility (Manhattan, KS)				8776	8776
Elite Elastomers Inc (Ripley, MS)	19.31			8532.47	8551.78
Univar USA Inc Romulus Branch (Romulus, MI)	0.48			8503.998	8504.478
Anamet Electrical Inc (Mattoon, IL)				7960	7960
Forbo Siegling LLC (Huntersville, NC)	162			7624	7786
GRT Rubber Technologies LLC (Paragould, AR)	325.34	40.84		7239	7605.18
Ford Motor Company - Kansas City Assembly Plant (Claycomo, MO)	2312			5179	7491
Natvar (Clayton, NC)	1750			5250	7000
Sumiriko Ohio Inc (Bluffton, OH)				6890	6890
Nexeo Solutions LLC (Twinsburg, OH)	744			5946	6690
Colorite Polymers (Ridgefield, NJ)				6400	6400
Cooper Standard (Boiling Springs, SC)				5902	5902
Natvar (City Of Industry, CA)	1250			3750	5000
Buzzi Unicem USA-Cape Girardeau (Cape Girardeau, MO)	168			4796	4964
Sioux Chief (Peculiar, MO)	39			4823	4862
Hallstar Co Memphis Solutions Facility (Memphis, TN)	2932			1894	4826
Ferro Electronic Material Systems (Vista, CA)	274			4027	4301
LG Hausys America Inc (Adairsville, GA)	2500			1603	4103
Pass %26 Seymour Legrand (Concord, NC)	2750		750		3500
Baxter Healthcare of Puerto Rico (Madison, NC)	3015				3015
Parker Hannifin Corp (Kennett, MO)				2914.1	2914.1
Lanxess Solutions US Inc (Gastonia, NC)	90			2733	2823
Rex-Hide Industries Inc (Grafton, WV)	525	270	500	1520	2815

American Renolit Corp LA (Commerce, CA)	2423	255	2678
Preferred Compounding Corp (Barberton, OH)		2650	2650
Cardinal Health (Jacksonville, TX)		2519	2519
O%27Sullivan Films Inc (Winchester, VA)	2478.43		2478.43
Nexeo Solutions LLC (Saint Louis, MO)	750	1553	2303
Parker Hannifin (McCook, NE)	23.44	2260	2283.44
Gold Key Processing Inc (Middlefield, OH)		750	1500
Powell Electrical Systems Inc (Houston, TX)	553	1660	2213
Nexeo Solutions LLC-Doraville (Doraville, GA)	356	1573	1929
Harwick Standard Distribution Corp (Akron, OH)	45	1867	1912
Akwel Cadillac USA Inc (Cadillac, MI)	1550.02	179.89	1729.91
Teel Plastics Inc (Baraboo, WI)		1692	1692
Basf Corp (Riverside Facility) (Riverside, CA)		1500	1500
Ash Grove Cement (Foreman, AR)	0.066	1374.31	1374.376
Ford Motor Co - Ohio Assembly Plant (Avon Lake, OH)		1310	1310
Cary Compounds LLC (Dayton, NJ)	249.56	911.59	1161.15
Emco Chemical Distributors Inc (Pleasant Prairie, WI)	255	862.46	1117.46
Veolia Es Technical Solutions LLC Port Arthur Facility (Port Arthur, TX)	20	1080	1100
BD Life Sciences (Sumter, SC)	1	1088	1089
Ennis-Flint Salem (Salem, OR)	376.06	677	1053.06
Eagle Packaging Inc (Earth City, MO)	21	962	983
Valley Processing (City Of Industry, CA)	957.18	0	957.18
Barton Solvents Inc Council Bluffs (Council Bluffs, IA)	5	835	840
Preformed Line Products Co (Rogers, AR)		750	750
Winchester Interconnect Cm Corp (Dayville, CT)		747.87	747.87
Brenntag Mid-South (Charlotte, NC)	2	668	670
Delta Rubber Co (Danielson, CT)	10.7	652	662.7

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Honda Of America Manufacturing Inc (East Liberty, OH)	10		630	640
Emco Chemical Distributors Inc (North Chicago, IL)	520		15	535
Henkel Us Operations Corp (Oak Creek, WI)	20		500	520
Cardinal Systems Inc (Schuylkill Haven, PA)	10		500	510
Ace Elastomer (Rock Hill, SC)			500	500
Heritage Thermal Services (East Liverpool, OH)	3.00935	0.17	479.4405	482.6199
Parker Hannifin Corp O-Ring Div (Lebanon, TN)	0.0075842		464.24	464.2476
Honda Of America Manufacturing Inc (Marysville, OH)	18		376	394
Univar USA Inc (Kent, WA)	357			357
Nov Rig Systems West Little York (Houston, TX)			336	336
Loxcreen Co Inc (Woodburn, OR)	242		72.4	314.4
Rex-Hide Industries Inc (Tyler, TX)	21	10	275	306
Continental Cement Co LLC (Hannibal, MO)	37.49		191.01	228.5
Mexichem Specialty Compounds (Pineville, NC)	39		167.34	206.34
LMI Custom Mixing LLC (Cambridge, OH)			185	185
Eaton Corp (Mountain Home, AR)	80.23	3	88	171.23
C%26M Corp (Wauregan, CT)	3		37	121
Axiall LLC (Plaquemine, LA)			146	146
Vytron Corp (Loudon, TN)			115	115
Polyone Corp (Terre Haute, IN)		25	79.2	104.2
Mitsubishi Motors Na Inc (Normal, IL)			72	72
Barton Solvents Inc Bettendorf (Bettendorf, IA)	0.001		68.7	68.701
Weir Slurry Group Inc (Madison, WI)	68.466			68.466
Mexichem Specialty Compounds (Leominster, MA)	60		5	65
Manner Polymers Inc (Mckinney, TX)	56.68		4.23	60.91
Buckhorn Rubber Products Inc (Hannibal, MO)	60.33			60.33
Giant Cement Co (Harleyville, SC)	5.91		48.72	54.63

Elpaco Coatings LLC (Pagedale, MO)	51				51
3M Cottage Grove Center (Cottage Grove, MN)		44			44
Barton Solvents Inc Kansas City (Kansas City, KS)	0.002		41.4		41.402
Vital Signs Inc (Englewood, CO)	20		15		35
Preferred Compounding Corp (Huntingdon, TN)			33		33
Akzo Nobel Functional Chemicals LLC (Pasadena, TX)			31		31
Pexco LLC (Athol, MA)	28				28
Hexpol Compounding Fdba Chase Elastomer A Hexpol Co (Kennedale, TX)			24		24
Hexpol Compounding Burton Rubber Processing (Jonesborough, TN)			15		15
Lehigh Cement Co (Logansport, IN)	11				11
Pexco LLC (Tacoma, WA)	9				9
Veolia Es Technical Solutions LLC (Sauget, IL)			6		6
Doremus Terminal LLC (Newark, NJ)	2				2
The Dow Chemical Co - Louisiana Operations (Plaquemine, LA)			1.03		1.03
Clean Harbors Deer Park LLC (La Porte, TX)	0.1				0.1
Ash Grove Cement Co (Chanute, KS)	0.03				0.03
Barnhardt Manufacturing Co NCFI Polyurethanes Div (Mount Airy, NC)	0.01118				0.01118
TOTAL	266,230.87	11,141	226,287	21,315,331	21,818,990

Source: table derived from EPA Toxics Release Inventory, November 2019.

Table 2. DEHP Waste Transfers, Pounds, 2012-2018

Destination	Facility Type (per EPA Chemical Data Exchange)	Pounds (2012-2018)
JP Plastics (Charlotte, NC)	Recycling: Transfer to Waste Broker	5323402.05
Baxter Sa De Cuernavaca (Jiutepec, Mor., Mexico)	Recycling: Other Reuse or Recovery	1912668.3
Corporacion Comunitaria Reciclaje Del Norte (Hatillo, PR)	Recycling: Transfer to Waste Broker	1208633.8
IFCO Recycling Inc (Caguas, PR)	Recycling: Other Reuse or Recovery	970439.01
Plasticos Del Caribe (Santo Domingo, Dominican Republic)	Recycling: Transfer to Waste Broker	947950
TM Deer Park (Deer Park, TX)	Disposal: Underground Injection to Class II-V Wells	693240
Tenex Corporation (Mountain Home, AR)	Recycling: Other Reuse or Recovery	642185
Courtney Ridge Landfill (Sugar Creek, MO)	Disposal: Other Landfills	635788
J.P. Plastics C/O Brunson Recycling (Marion, NC)	Recycling: Other Reuse or Recovery	612039
Waste Management Of New York, LLC (Chaffee, NY)	Disposal: Other Landfills	583977
CJ Poly (Plymouth, IN)	Recycling: Transfer to Waste Broker	570488.83
Ashland Chemical Co (Binghamton, NY)	Treatment: Incineration/Thermal Treatment	407780.07
Heritage Environmental Services LLC (Indianapolis, IN)	Treatment: Solidification/Stabilization	374749.7
Foothills Environmental Landfill (Lenoir, NC)	Disposal: Other Landfills	326141.15
Quincy Recycle (Alsip, IL)	Recycling: Transfer to Waste Broker	298034.94
Us Ecology Texas Inc (Robstown, TX)	Disposal: Other Off-Site Management	284600.99
Veolia Es Technical Solutions (Port Arthur, TX)	Treatment: Incineration/Thermal Treatment	276867
Northeast Ne Solid Waste Coalition Landfill (Clarkson, NE)	Disposal: Other Landfills	269774.4
Clean Harbors (Braintree, MA)	Energy Recovery: Transfer to Waste Broker	266836
Waste Management Prairie Bluff Landfill (Houston, MS)	Disposal: Other Landfills	236678
S. Slesnick Company (Canton, OH)	Recycling: Other Reuse or Recovery	230461
Safety Kleen Inc (Springfield, MO)	Recycling: Transfer to Waste Broker	229297
Samar Incorporated (Stoughton, MA)	Recycling: Transfer to Waste Broker	197620

Vish Group (North Brunswick, NJ)	Recycling: Transfer to Waste Broker	188187
Georgia-Pacific Chemicals LLC (Millersburg, OR)	Treatment: Incineration/Thermal Treatment	187885
Nexeo Solutions LLC (Binghamton, NY)	Treatment: Solidification/Stabilization	163099
Recycling Solutions, Inc. (Arecibo, PR)	Recycling: Transfer to Waste Broker	150679.802
Bristol Va Landfill (Bristol, VA)	Disposal: Other Landfills	148113
Custom Polymers (Charlotte, NC)	Recycling: Transfer to Waste Broker	145167
Celina Landfill Inc * (Celina, OH)	Disposal: Transfer to Waste Broker - Disposal	144595
Greenway Recover %26 Recycle LLC (Union City, TN)	Recycling: Transfer to Waste Broker	130455
Haywood County Landfill (Brownsville, TN)	Disposal: Other Landfills	117275
Lakeland Polymers LLC (Boonton, NJ)	Recycling: Other Reuse or Recovery	106715.345
Lagrange City Landfill (Lagrange, GA)	Disposal: Other Landfills	106316
Hahnco Plastics Inc. (Riverside, CA)	Recycling: Transfer to Waste Broker	106251
Clean Harbors Environmental Services, Inc. (Kimball, NE)	Treatment: Solidification/Stabilization	97439
Kimble Recycling Disposal Facility (Dover, OH)	Disposal: Other Landfills	86200
Siouxland Mat Company (Sioux Center, IA)	Recycling: Other Reuse or Recovery	76281.88
K%26B Plastic Industries (Blodgett Mills, NY)	Recycling: Other Reuse or Recovery	73376.2
Backridge Landfill (La Grange, MO)	Disposal: Other Landfills	71135
Coll Materials (Allentown, PA)	Recycling: Transfer to Waste Broker	67683
Covanta Energy (Indianapolis, IN)	Energy Recovery: Transfer to Waste Broker	60161.42
Countywide Rfd (East Sparta, OH)	Disposal: Other Landfills	56573
Republic Services - Union County Regional (Enoree, SC)	Disposal: Other Landfills	52002
Bluff Road Landfill (Lincoln, NE)	Disposal: Other Landfills	46228.5231
Mumford Industries (Ninety-Six, SC)	Recycling: Other Reuse or Recovery	44212
Prairie View Landfill (Wyatt, IN)	Disposal: Other Land Disposal	44091
Great River Regional Landfill (Fort Madison, IA)	Disposal: Other Landfills	43775
Northwest Tennessee Disposal (Union City, TN)	Disposal: Other Landfills	43685.73
Dimex LLC (Marietta, OH)	Recycling: Other Reuse or Recovery	41245.836

Tri-State Metal Co (Chicago, IL)	Recycling: Other Reuse or Recovery	39118
Us Wire and Cable (Newark, NJ)	Recycling: Transfer to Waste Broker	37670.035
Us Ecology Nevada Inc (Beatty, NV)	Disposal: RCRA Subtitle C Landfills	36986
Lake View LDFL (Erie, PA)	Disposal: Other Landfills	36700
Itasca Landfill (Itasca, TX)	Disposal: Management Method Unknown	35616
Evergreen Recycling %26 Disposal (Northwood, OH)	Disposal: Other Landfills	34650
Tradebe Treatment and Recycling Of Tennessee, LLC (Millington, TN)	Energy Recovery: Energy Recovery	32622
Vls - Armor, LLC (Mount Pleasant, TN)	Treatment: Incineration/Insignificant Fuel Value	30878
Wexford County Landfill (Manton, MI)	Disposal: Other Landfills	30627.69
Lopez Scrap Metal (El Paso, TX)	Recycling: Other Reuse or Recovery	27682
Systech Environmental Corp. (Fredonia, KS)	Energy Recovery: Energy Recovery	27056.3
Giant Resource Recovery-Sumter, Inc. (Sumter, SC)	Energy Recovery: Energy Recovery	26772.72
J Bar J Landfill Inc (Ogallala, NE)	Disposal: RCRA Subtitle C Landfills	26242
Clean Harbors (Cincinnati, OH)	Treatment: Incineration/Thermal Treatment	25849.03
Tradebe Treatment %26 Recycling, LLC (East Chicago, IN)	Energy Recovery: Transfer to Waste Broker	25098.5599
Broadhurst Environmental (Screven, GA)	Disposal: Other Landfills	24589
Sparton Enterprises, Inc. (Norton, OH)	Recycling: Other Reuse or Recovery	24210
Resourceful Environmental Services (Ripley, MS)	Disposal: Other Landfills	23399.86
Environtech Inc (Morris, IL)	Disposal: Other Land Disposal	21601
Waste Management - Pheasant Run RDF (Bristol, WI)	Disposal: Other Landfills	20162
Kimble Recycling Services (Twinsburg, OH)	Disposal: Other Landfills	19667.6
EQ Detroit Inc (Detroit, MI)	Energy Recovery: Energy Recovery	19186.4
Msw Plastics Inc. (Ponce, PR)	Recycling: Other Reuse or Recovery	18911.934
Republic - North Shelby (Millington, TN)	Disposal: Other Landfills	18541
Huntsville Solid Waste Disposal Authority (Huntsville, AL)	Energy Recovery: Energy Recovery	18016
Deruijter International USA, Inc (Coldwater, OH)	Recycling: Other Reuse or Recovery	17036.84
Lancaster Cnty Swma Transf Sta (Lancaster, PA)	Disposal: Other Land Disposal	16776

American Recycling Center, Inc. (Owosso, MI)	Recycling: Transfer to Waste Broker	16485.19
Palmetto Landfill (Spartanburg, SC)	Disposal: Other Landfills	15597
Jebro Inc (Sioux City, IA)	Energy Recovery: Energy Recovery	15541.8029
Security Rdf (Cleveland, TX)	Disposal: Other Land Disposal	15237.31
Safety-Kleen Systems (Dolton, IL)	Energy Recovery: Energy Recovery	14212.87
C4 Polymers, Inc (Chagrin Falls, OH)	Recycling: Other Reuse or Recovery	14119
J.P. Industries (Lisbon, OH)	Recycling: Other Reuse or Recovery	13749
N/A	POTW Treatment: Transfer To Publicly Owned Treatment Works	13703.6663
Ocoya Landfill (Pontiac, IL)	Disposal: Other Land Disposal	13223
Lcswma Frey Farm Landfill (Conestoga, PA)	Disposal: Other Landfills	13218
Ag Polymers (White Plains, NY)	Recycling: Other Reuse or Recovery	13125
American Environmental Svcs (Calvert City, KY)	Energy Recovery: Energy Recovery	12998.896
Scott Solid Waste Disposal Company (Oneida, TN)	Disposal: Other Landfills	12288
Vopak Industrial Serv (Mauldin, SC)	Energy Recovery: Energy Recovery	11884
Phillip Reclamation Services (Houston, TX)	Disposal: Transfer to Waste Broker - Disposal	11385
Lancaster Cnty Swa Trash Incinerator (Bainbridge, PA)	Energy Recovery: Energy Recovery	10888.754
Industry City Of Dspl Co (City Of Industry, CA)	Disposal: Other Landfills	10253
Cherokee Run Landfill (Bellefontaine, OH)	Disposal: Other Landfills	9386
Eagle Plastics (Dover, NJ)	Recycling: Other Reuse or Recovery	9000
Topeka Waste Sys (Topeka, KS)	Disposal: Other Landfills	8776
Iowa Waste Systems Inc. (Red Oak, IA)	Disposal: Other Landfills	8664
Clean Harbors Aragonite LLC (Grantsville, UT)	Treatment: Incineration/Thermal Treatment	7989.1
Schoenberg %26 Co., Inc. (Yorktown, NY)	Recycling: Transfer to Waste Broker	7960
Waste Management Of SC (Wellford, SC)	Disposal: Other Landfills	7297
Northeast Arkansas Regional Swmd Landfill (Paragould, AR)	Disposal: Other Landfills	7239
Rineco Chemical (Benton, AR)	Energy Recovery: Energy Recovery	7127.54
Spring Grove Resource Recovery (Cincinnati, OH)	Energy Recovery: Energy Recovery	7076

Safety-Kleen Systems Inc (Smithfield, KY)	Energy Recovery: Transfer to Waste Broker	6654.844
Republic Services (Greenville, SC)	Disposal: Other Landfills	5902
Polymeric Inc (Cuyahoga Falls, OH)	Recycling: Other Reuse or Recovery	5500
Heritage-Wti Inc. (East Liverpool, OH)	Treatment: Incineration/Insignificant Fuel Value	5418
Browning-Ferris Ind Of Ca Inc (Sylmar, CA)	Disposal: Other Landfills	5400
Tier Environmental LLC DBA Hukill Environmental Service (Bedford, OH)	Treatment: Incineration/Insignificant Fuel Value	5286
Johnston Co Landfill (Smithfield, NC)	Disposal: Other Landfills	5250
Axion Recycled Plastics Inc. (Zanesville, OH)	Recycling: Transfer to Waste Broker	5075
Michigan Marketing (Portage, MI)	Recycling: Transfer to Waste Broker	5051.65
Summit Waste Systems (Lee%27S Summit, MO)	Disposal: Other Landfills	4823
Cherokee Sanitary Landfill (Cherokee Village, AR)	Disposal: Other Landfills	4750
Chemtron Corp (Avon, OH)	Disposal: Transfer to Waste Broker - Disposal	4718.5
Ecoflo, Inc (Greensboro, NC)	Energy Recovery: Energy Recovery	4459
Waste Management - Metro Rdf (Franklin, WI)	Disposal: Other Landfills	4324
Ecm (Union City, TN)	Disposal: Other Landfills	4257
Orchard Enterprises (Hanover, MA)	Recycling: Other Reuse or Recovery	3869
Teknor Apex Tennessee Company (Aka Haywood Co) (Brownsville, TN)	Recycling: Other Reuse or Recovery	3791
Valley Vista Services (City of Industry, CA)	Disposal: Other Landfills	3750
Johnson County Landfill (Shawnee, KS)	Disposal: Other Off-Site Management	3730
Advanced Waste Services (Sheboygan, WI)	Disposal: Other Landfills	3672
Cntrl Landfill Resource Recovery Plant (Johnston, RI)	Disposal: Other Landfills	3592
Resource One (Cincinnati, OH)	Recycling: Transfer to Waste Broker	3508
Heritage-Crystal Clean (Elgin, IL)	Recycling: Other Reuse or Recovery	3258
Veolia Es Technical Solutions LLC (West Carrollton, OH)	Disposal: Transfer to Waste Broker - Disposal	3255.2
Cycle Chem In (Lewisberry, PA)	Treatment: Incineration/Thermal Treatment	3163
Cowart Industrial Services (Carrollton, GA)	Energy Recovery: Transfer to Waste Broker	2890

Industrial Container Services Nc LLC (Charlotte, NC)	Treatment: Incineration/Thermal Treatment	2797.8
Lemons Landfill (Dexter, MO)	Disposal: Other Landfills	2788.3
Toro Energy of Ohio - America%27S Landfill Gas (Waynesburg, OH)	Disposal: Other Landfills	2650
Waste Management (Chardon, OH)	Disposal: Other Landfills	2639
Clean Harbors El Dorado LLC (El Dorado, AR)	Energy Recovery: Energy Recovery	2348
County Environmental Wyandot Landfill (Carey, OH)	Disposal: Other Landfills	2160
Barton Solvents Inc Council Bluffs (Council Bluffs, IA)	Recycling: Other Reuse or Recovery	2042
B. Schoenberg %26 Co (Arlington Heights, IL)	Recycling: Other Reuse or Recovery	2017
Adams Plastics Lp (Rolling Meadows, IL)	Recycling: Transfer to Waste Broker	2013
Cape Metal Recycling (Cape Girardeau, MO)	Disposal: Management Method Unknown	1985
American Landfill (Waynesburg, OH)	Disposal: Other Landfills	1944
Univar USA Inc Memphis (Memphis, TN)	Recycling: Other Reuse or Recovery	1894
Northwestern Landfill (Parkersburg, WV)	Disposal: Other Landfills	1782.1
501 (Bruceton, TN)	Recycling: Other Reuse or Recovery	1768.24
Columbia Cty Solid Waste Dept (Pardeeville, WI)	Disposal: Other Landfills	1692
Veolia Es Technical Solutions LLC (Sauget, IL)	Treatment: Incineration/Thermal Treatment	1632.8
Peoria Disposal Company (Peoria, IL)	Disposal: RCRA Subtitle C Landfills	1567
Lafarge Corporation (Sugar Creek, MI)	Energy Recovery: Energy Recovery	1461.755
Clean Harbors Deer Park, L.P. (La Porte, TX)	Treatment: Incineration/Thermal Treatment	1412
Memphis Drum Service Inc (Memphis, TN)	Treatment: Other Waste Treatment	1387
Ash Grove Cement Co (Chanute, KS)	Energy Recovery: Energy Recovery	1374.31
Fielding Environmental (Mississauga, ON)	Energy Recovery: Energy Recovery	1368
Buzzi Unicem USA Greencastle Alternate Fuels (Greencastle, IN)	Energy Recovery: Energy Recovery	1314
BFI of Ohio Inc (Oberlin, OH)	Disposal: Other Landfills	1310
Meadowfill Landfill (Bridgeport, WV)	Disposal: RCRA Subtitle C Landfills	1250
Clean Harbors Reidsville Limited Liability Company (Reidsville, NC)	Energy Recovery: Energy Recovery	1194

Bruckman Rubber Co (Hastings, NE)	Recycling: Other Reuse or Recovery	1167.34
Atco-Mission Industrial Inc (Houston, TX)	Recycling: Transfer to Waste Broker	1160
Avangard Innovative (Houston, TX)	Recycling: Transfer to Waste Broker	1159
Pas Rubber, LLC (Washington Twp, MI)	Recycling: Other Reuse or Recovery	1069.69
United Solutions, Inc. (Franklin, OH)	Treatment: Wastewater Treatment (Excluding POTW)	1062
Allstate Orc (West Milford, NJ)	Disposal: Transfer to Waste Broker - Disposal	1000
Glenwillow Sanitary Landfill * (Glenwillow, OH)	Disposal: Other Landfills	972
Sugar Creek Cement Plant (Sugar Creek, MO)	Energy Recovery: Energy Recovery	924.29
Frontier Logistical Services LLC (Memphis, TN)	Recycling: Other Reuse or Recovery	923.01
Waste Management Of Pasadena (Pasadena, TX)	Disposal: Other Landfills	910
Giant Resource Recovery- Attalla, Inc. (Attalla, AL)	Treatment: Incineration/Insignificant Fuel Value	878.17
Prime Materials Recovery Inc (Willimantic, CT)	Disposal: Transfer to Waste Broker - Disposal	868.87
Midwest Elastomers Inc (Wapakoneta, OH)	Recycling: Transfer to Waste Broker	800
Twin City Container Inc (Hastings, MN)	Treatment: Other Waste Treatment	753.4
Proler Southwest (Houston, TX)	Disposal: Other Landfills	750
Emil A Schroth Inc (Howell, NJ)	Disposal: Other Off-Site Management	750
Waste Management Tontitown Landfill, LLC(Springdale, AR)	Disposal: Other Landfills	750
Midco Waste (New Brunswick, NJ)	Disposal: Other Landfills	722
Universal Composites (Bay Shore, NY)	Recycling: Transfer to Waste Broker	706.33
Bahan Machine %26 Foundry (Greenville, SC)	Treatment: Other Waste Treatment	632
Green America Recycling LLC (Hannibal, MO)	Energy Recovery: Energy Recovery	591
Eagle Environmental Services Dorchester Facility (Dorchester, SC)	Disposal: Other Land Disposal	544
Lancaster Oil Company (Lancaster, PA)	Treatment: Solidification/Stabilization	543
Bfi-South Shelby Landfill (Memphis, TN)	Disposal: Transfer to Waste Broker - Disposal	507
Commonwealth Environmental Systems (Hegins, PA)	Disposal: Other Landfills	505
Cabarrus County Landfill (Concord, NC)	Disposal: Transfer to Waste Broker - Disposal	500

Wastewater Treatment Plant (Hannibal, MO)	POTW Treatment: Transfer To Publicly Owned Treatment Works	500
Veolia Es Emerald Park Landfill (Muskego, WI)	Disposal: Other Land Disposal	500
Freedom Waste Landfill (Mayfield, KY)	Disposal: Other Landfills	484
Heffco Elastomers Inc. (Simcoe, ON)	Recycling: Transfer to Waste Broker	480
Cone Solvents Inc (Memphis, TN)	Recycling: Other Reuse or Recovery	404
Michigan Disposal Waste Treatment Plant (Belleville, MI)	Energy Recovery: Energy Recovery	383.51
Henkel Corp (Oak Creek, WI)	Disposal: Management Method Unknown	340
Valicor Environmental Services LLC (Middletown, OH)	Treatment: Wastewater Treatment (Excluding POTW)	335
Veolia Es Arbor Hills Landfill (Northville, MI)	Disposal: Other Landfills	285.09
Lee County Landfill Sc LLC (Bishopville, SC)	Disposal: RCRA Subtitle C Landfills	272
Kiln Direct Inc (Sumter, SC)	Disposal: Other Land Disposal	272
Redline Metals, Inc (Lombard, IL)	Recycling: Other Reuse or Recovery	263
East Texas Waste Management (Tyler, TX)	Disposal: RCRA Subtitle C Landfills	260
Grafton City Of (Grafton, WV)	POTW Treatment: Transfer To Publicly Owned Treatment Works	255
USA Environmental Incorporated (Detroit, MI)	Disposal: Other Landfills	250
Aa Sydcot LLC (Yuma, AZ)	Disposal: Transfer to Waste Broker - Disposal	250
Olympic Waste (Jacksonville, TX)	Disposal: Other Landfills	200
Cadillac WWTP (Cadillac, MI)	POTW Treatment: Transfer To Publicly Owned Treatment Works	179.89
Veolia Es Technical Solutions Corp (Flanders, NJ)	Disposal: Management Method Unknown	174.04
WRR Environmental Services Co Inc (Eau Claire, WI)	Energy Recovery: Energy Recovery	164.1
Parma Custom Cryogenics (Waterford, ON)	Recycling: Transfer to Waste Broker	152
Enterprise Oil (Knoxville, TN)	Recycling: Other Reuse or Recovery	145
West Camden Sanitary Landfill (Camden, TN)	Disposal: Other Landfills	130.4
ECM Of Ridgely, LLC (Ridgely, TN)	Disposal: Other Landfills	125.8
Mcalpine Creek WWTP (Pineville, NC)	POTW Treatment: Transfer To Publicly Owned Treatment Works	125.14

High Point Materials (Milford, PA)	Recycling: Other Reuse or Recovery	117.12
Loudon County Landfill (Loudon, TN)	Disposal: Other Landfills	115
Cedar Ridge Landfill (Lewisburg, TN)	Disposal: Other Landfills	104.72
Marchem Southeast (Adairsville, GA)	Recycling: Other Reuse or Recovery	81
Coyne Textile Services (Cleveland, OH)	Energy Recovery: Transfer to Waste Broker	75
Covanta Marion Energy Recovery Facility (Brooks, OR)	Energy Recovery: Energy Recovery	72.4
Chemical Waste Management of The Nw (Arlington, OR)	Treatment: Incineration/Insignificant Fuel Value	55
Area Disposal Services Inc (Clinton, IL)	Disposal: Other Landfills	46
City of Dover Wastewater Treatment Plant (Dover, OH)	POTW Treatment: Transfer To Publicly Owned Treatment Works	35
Wayne Disposal Inc (Belleville, MI)	Disposal: RCRA Subtitle C Landfills	27.7
Mid-America Steel Drum Co In C (Oak Creek, WI)	Treatment: Other Waste Treatment	27.6
Stablex Canada Inc. (Blainville, ON)	Treatment: Solidification/Stabilization	26
Dfw Waste Oil Service (Alvarado, TX)	Energy Recovery: Energy Recovery	24
Us Ecology Michigan Disposal (Belleville, MI)	Disposal: RCRA Subtitle C Landfills	22
Univar USA Inc Houston Fm 529 (Houston, TX)	Disposal: Management Method Unknown	21
Meridian Metropolitan District (Englewood, CO)	POTW Treatment: Transfer To Publicly Owned Treatment Works	15
City Of Grafton WWTP (Grafton, WV)	POTW Treatment: Transfer To Publicly Owned Treatment Works	15
North Shore Sanitary District (Gurnee, IL)	POTW Treatment: Transfer To Publicly Owned Treatment Works	15
Appalachian Environmental Resources, Inc. (Johnson City, TN)	Energy Recovery: Energy Recovery	15
City Of Tyler- Westside Wastewater Treatment Plant (Tyler, TX)	POTW Treatment: Transfer To Publicly Owned Treatment Works	15
Regen Inc. (Middlebury, IN)	Recycling: Other Reuse or Recovery	14.685
Indianapolis Drum (Indianapolis, IN)	Disposal: Transfer to Waste Broker - Disposal	14.4
Ross Incineration Services Inc (Grafton, OH)	Treatment: Incineration/Thermal Treatment	11.6
Jones Barrel Co (Omaha, NE)	Disposal: Management Method Unknown	10.09

Mountain Home Wastewater Treatment Plant (Mountain Home, AR)	POTW Treatment: Transfer To Publicly Owned Treatment Works	10
South Huron Valley WWT (Rockwood, MI)	POTW Treatment: Transfer To Publicly Owned Treatment Works	8
Safety Kleen Systems Inc (South Plainfield, NJ)	Disposal: Management Method Unknown	7.42
Methvin Sanitation Inc. (Harrison, AR)	Disposal: Other Landfills	7
Cambridge Water Pollution Control Center (Cambridge, OH)	POTW Treatment: Transfer To Publicly Owned Treatment Works	5
Town of Killingly Water Pollution Control Facility (Danielson, CT)	POTW Treatment: Transfer To Publicly Owned Treatment Works	5
Excelsior Springs Water Treatment Plant (Excelsior Springs, MO)	POTW Treatment: Transfer To Publicly Owned Treatment Works	5
Passaic Valley Sewer Comm (Newark, NJ)	POTW Treatment: Transfer To Publicly Owned Treatment Works	5
Clark Steel Drum (Peabody, MA)	Disposal: Management Method Unknown	5
Vickery Environmental Inc (Vickery, OH)	Disposal: Underground Injection to Class I Wells	5
San Jose Creek Water Reclamation Plant (Whittier, CA)	POTW Treatment: Transfer To Publicly Owned Treatment Works	5
Ntmwd 121 Regional Disposal Facility (Melissa, TX)	Disposal: Other Landfills	4.23
Milam Recycling %26 Disposal (East St Louis, IL)	Disposal: Other Landfills	3
Consolidated Container Corp (Minneapolis, MN)	Treatment: Other Waste Treatment	3
TOTAL		21,314,838.61

Source: U.S. EPA Chemical Data Exchange.

ENDNOTES

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